SIMATS ENGINEERING

ASSESSMENT TOOL 1 – High (Coding Challenge)

COURSE CODE: CSA0904

COURSE NAME: Programming in Java

NAME: K. ANJALI GIRI

REG NO: 192421037

1. Library Book Management using Classes & Encapsulation

```
import java.util.Scanner;

class Book {
    private int id;
    private String title, author;
    private double price;
    private boolean issued;

    Book(int id, String title, String author, double price) {
        this.id = id;
        this.title = title;
        this.author = author;
        this.price = price;
    }

    int getId() {
        return id;
    }

    void issueBook() {
        if (!issued) {
            issued = true;
            System.out.println("Book Issued");
        } else
            System.out.println("Already Issued");
    }

    void returnBook() {
        if (issued) {
            issued = false;
            System.out.println("Book Returned");
        } else
```

```java
            System.out.println("Book Not Issued");
    }

    void display() {
        System.out.println(id + " " + title + " " + author + " Rs." + price);
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        Book[] b = new Book[5];
        int count = 0, choice, id;

        do {
            System.out.println("\n1.Add  2.Display  3.Issue  4.Return  5.Search  6.Exit");
            System.out.print("Choice: ");
            choice = sc.nextInt();

            switch (choice) {

                case 1:
                    sc.nextLine();
                    System.out.print("Book ID: ");
                    id = sc.nextInt();
                    sc.nextLine();
                    System.out.print("Title: ");
                    String t = sc.nextLine();
                    System.out.print("Author: ");
                    String a = sc.nextLine();
                    System.out.print("Price: ");
                    double p = sc.nextDouble();
                    b[count++] = new Book(id, t, a, p);
                    System.out.println("Book Added");
                    break;

                case 2:
                    for (int i = 0; i < count; i++)
                        b[i].display();
                    break;

                case 3:
                    System.out.print("Book ID: ");
```

```
                id = sc.nextInt();
                for (int i = 0; i < count; i++)
                    if (b[i].getId() == id)
                        b[i].issueBook();
                break;

            case 4:
                System.out.print("Book ID: ");
                id = sc.nextInt();
                for (int i = 0; i < count; i++)
                    if (b[i].getId() == id)
                        b[i].returnBook();
                break;

            case 5:
                System.out.print("Book ID: ");
                id = sc.nextInt();
                boolean found = false;
                for (int i = 0; i < count; i++)
                    if (b[i].getId() == id) {
                        b[i].display();
                        found = true;
                    }
                if (!found)
                    System.out.println("Book Not Found");
                break;

            case 6:
                System.out.println("Thank You");
        }

    } while (choice != 6);

    sc.close();
  }
}
```

Output

```
1.Add  2.Display  3.Issue  4.Return  5.Search  6.Exit
Choice: 1
Book ID: 901
Title: JAVA
Author: CHARLES
Price: 500
Book Added

1.Add  2.Display  3.Issue  4.Return  5.Search  6.Exit
Choice: 2
901 JAVA CHARLES Rs.500.0

1.Add  2.Display  3.Issue  4.Return  5.Search  6.Exit
Choice: 3
Book ID: 900

1.Add  2.Display  3.Issue  4.Return  5.Search  6.Exit
Choice: 5
Book ID: 901
901 JAVA CHARLES Rs.500.0

1.Add  2.Display  3.Issue  4.Return  5.Search  6.Exit
Choice: 6
Thank You
```

2. Employee Payroll System using Inheritance

Program

```
class Employee {
    int id;
    String name;

    Employee(int id, String name) {
        this.id = id;
        this.name = name;
    }

    double calculateSalary() {
        return 0;
    }

    void display() {
        System.out.println("ID : " + id);
        System.out.println("Name : " + name);
        System.out.println("Salary : " + calculateSalary());
        System.out.println();
    }
}

class PermanentEmployee extends Employee {
    double basic, bonus;

    PermanentEmployee(int id, String name, double basic, double bonus) {
        super(id, name);
        this.basic = basic;
        this.bonus = bonus;
    }

    double calculateSalary() {
        return basic + bonus;
    }
}

class ContractEmployee extends Employee {
    int days;
    double pay;

    ContractEmployee(int id, String name, int days, double pay) {
        super(id, name);
```

```
        this.days = days;
        this.pay = pay;
    }

    double calculateSalary() {
        return days * pay;
    }
}

public class Main {
    public static void main(String[] args) {

        Employee[] emp = {
            new PermanentEmployee(101, "Anjali", 30000, 5000),
            new ContractEmployee(102, "Rahul", 25, 800)
        };

        for (Employee e : emp)
            e.display();
    }
}
```

Output

```
ID : 101
Name : Anjali
Salary : 35000.0

ID : 102
Name : Aishma
Salary : 20000.0


=== Code Execution Successful ===
```

3. Vehicle Rental System using Runtime Polymorphism

Program

```
import java.util.Scanner;

class Vehicle {
    String vehicleName;

    Vehicle(String vehicleName) {
        this.vehicleName = vehicleName;
    }

    double calculateRent(int days) {
        return 0;
    }

    void display(int days) {
        System.out.println("Vehicle : " + vehicleName);
        System.out.println("Days    : " + days);
        System.out.println("Rent    : Rs." + calculateRent(days));
        System.out.println();
    }
}

class Car extends Vehicle {

    Car() {
        super("Car");
    }

    double calculateRent(int days) {
        return days * 1000;
    }
}

class Bike extends Vehicle {

    Bike() {
        super("Bike");
    }

    double calculateRent(int days) {
        return days * 500;
    }
}

class Bus extends Vehicle {

    Bus() {
        super("Bus");
```

```
    }

    double calculateRent(int days) {
        return days * 2000;
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Vehicle[] vehicles = {
                new Car(),
                new Bike(),
                new Bus()
        };

        System.out.print("Enter Rental Days: ");
        int days = sc.nextInt();

        System.out.println("\nRental Bill");

        for (Vehicle v : vehicles)
            v.display(days);

        sc.close();
    }
}
```

Output

```
Enter Rental Days: 5

Rental Bill
Vehicle : Car
Days    : 5
Rent    : Rs.5000.0

Vehicle : Bike
Days    : 5
Rent    : Rs.2500.0

Vehicle : Bus
Days    : 5
Rent    : Rs.10000.0
```

4. Bank Account System using Data Hiding
Program

```
import java.util.Scanner;

class BankAccount {
    private int accountNumber;
    private String holderName;
    private double balance;

    // Constructor
    BankAccount(int accountNumber, String holderName, double balance) {
        this.accountNumber = accountNumber;
        this.holderName = holderName;
        this.balance = balance;
    }

    // Deposit
    void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.println("Amount Deposited Successfully");
        } else {
            System.out.println("Invalid Deposit Amount");
        }
    }

    // Withdraw
    void withdraw(double amount) {
        if (amount <= balance) {
            balance -= amount;
            System.out.println("Amount Withdrawn Successfully");
        } else {
            System.out.println("Insufficient Balance");
        }
    }

    // Balance Inquiry
    void showBalance() {
        System.out.println("\nAccount Number : " + accountNumber);
        System.out.println("Holder Name    : " + holderName);
        System.out.println("Balance        : Rs." + balance);
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Account Number: ");
```

```java
        int accNo = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Holder Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Initial Balance: ");
        double balance = sc.nextDouble();

        BankAccount account = new BankAccount(accNo, name, balance);

        System.out.print("\nEnter Deposit Amount: ");
        account.deposit(sc.nextDouble());

        System.out.print("Enter Withdrawal Amount: ");
        account.withdraw(sc.nextDouble());

        account.showBalance();

        sc.close();
    }
}
```

Output

```
Enter Account Number: 105128
Enter Holder Name: anjali
Enter Initial Balance: 6000

Enter Deposit Amount: 500
Amount Deposited Successfully
Enter Withdrawal Amount: 5000
Amount Withdrawn Successfully

Account Number : 105128
Holder Name    : anjali
Balance        : Rs.1500.0

=== Code Execution Successful ===
```

5. University Admission System using Method Overloading

Program

```
import java.util.Scanner;

class Admission {

    // Undergraduate Fee
    void calculateFee() {
        System.out.println("Undergraduate Fee : Rs.50000");
    }

    // Postgraduate Fee
    void calculateFee(int pg) {
        System.out.println("Postgraduate Fee : Rs.70000");
    }

    // Scholarship Fee
    void calculateFee(double scholarship) {
        System.out.println("Scholarship Fee : Rs.20000");
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        Admission a = new Admission();

        System.out.println("1. Undergraduate");
        System.out.println("2. Postgraduate");
        System.out.println("3. Scholarship");
        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();

        switch (choice) {

            case 1:
                a.calculateFee();
                break;

            case 2:
                a.calculateFee(1);
                break;

            case 3:
                a.calculateFee(1.0);
                break;

            default:
```

```
            System.out.println("Invalid Choice");
        }

        sc.close();
    }
}
```

Output

```
1. Undergraduate
2. Postgraduate
3. Scholarship
Enter Choice: 2
Postgraduate Fee : Rs.70000

=== Code Execution Successful ===
```

6. Shape Area Calculator using Abstract Class & Polymorphism

Program

```
import java.util.Scanner;

abstract class Shape {
    abstract void area();
}

class Circle extends Shape {
    double r;

    Circle(double r) {
        this.r = r;
    }

    void area() {
        System.out.println("Circle Area = " + (3.14 * r * r));
    }
}

class Rectangle extends Shape {
    double l, b;

    Rectangle(double l, double b) {
        this.l = l;
        this.b = b;
    }

    void area() {
        System.out.println("Rectangle Area = " + (l * b));
    }
}

class Triangle extends Shape {
    double base, height;

    Triangle(double base, double height) {
        this.base = base;
        this.height = height;
    }

    void area() {
        System.out.println("Triangle Area = " + (0.5 * base * height));
    }
}

public class Main {
    public static void main(String[] args) {
```

```java
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Circle Radius: ");
        double r = sc.nextDouble();

        System.out.print("Enter Rectangle Length: ");
        double l = sc.nextDouble();

        System.out.print("Enter Rectangle Breadth: ");
        double b = sc.nextDouble();

        System.out.print("Enter Triangle Base: ");
        double base = sc.nextDouble();

        System.out.print("Enter Triangle Height: ");
        double h = sc.nextDouble();

        Shape[] s = new Shape[3];

        s[0] = new Circle(r);
        s[1] = new Rectangle(l, b);
        s[2] = new Triangle(base, h);

        System.out.println("\nArea of Shapes");

        for (Shape shape : s)
            shape.area();

        sc.close();
    }
}
```

Output

```
Enter Circle Radius: 5
Enter Rectangle Length: 6
Enter Rectangle Breadth: 8
Enter Triangle Base: 4
Enter Triangle Height: 2

Area of Shapes
Circle Area = 78.5
Rectangle Area = 48.0
Triangle Area = 4.0

=== Code Execution Successful ===
```

7. Hospital Management System using Interfaces

Program

```
import java.util.Scanner;

interface MedicalRecord {
   void addRecord();
   void displayRecord();
}

class Patient implements MedicalRecord {
   int id;
   String name;
   Scanner sc = new Scanner(System.in);

   public void addRecord() {
      System.out.println("Enter Patient Details");
      System.out.print("Patient ID: ");
      id = sc.nextInt();
      sc.nextLine();
      System.out.print("Patient Name: ");
      name = sc.nextLine();
   }

   public void displayRecord() {
      System.out.println("\nPatient Record");
      System.out.println("ID : " + id);
      System.out.println("Name : " + name);
   }
}

class Doctor implements MedicalRecord {
   int id;
   String name;
   Scanner sc = new Scanner(System.in);

   public void addRecord() {
      System.out.println("\nEnter Doctor Details");
      System.out.print("Doctor ID: ");
      id = sc.nextInt();
      sc.nextLine();
      System.out.print("Doctor Name: ");
      name = sc.nextLine();
   }

   public void displayRecord() {
      System.out.println("\nDoctor Record");
      System.out.println("ID : " + id);
      System.out.println("Name : " + name);
   }
```

```
}

public class Main {
   public static void main(String[] args) {

      MedicalRecord[] records = new MedicalRecord[2];

      records[0] = new Patient();
      records[1] = new Doctor();

      for (MedicalRecord r : records)
         r.addRecord();

      System.out.println("\n----- Medical Records -----");

      for (MedicalRecord r : records)
         r.displayRecord();
   }
}
```

Output

```
Enter Patient Details
Patient ID: 201
Patient Name: devi

Enter Doctor Details
Doctor ID: 607
Doctor Name: farhana

----- Medical Records -----

Patient Record
ID : 201
Name : devi

Doctor Record
ID : 607
Name : farhana

=== Code Execution Successful ===
```

8. Online Shopping Order System using Packages

Program

```
import java.util.Scanner;

class Product {
    String name;
    double price;

    Product(String name, double price) {
        this.name = name;
        this.price = price;
    }
}

class Order {
    Product p;
    int quantity;

    Order(Product p, int quantity) {
        this.p = p;
        this.quantity = quantity;
    }

    void displayBill() {
        double total = p.price * quantity;

        System.out.println("\nOrder Details");
        System.out.println("Product Name : " + p.name);
        System.out.println("Price : " + p.price);
        System.out.println("Quantity : " + quantity);
        System.out.println("Total Amount : Rs." + total);
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Product Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Product Price: ");
        double price = sc.nextDouble();

        System.out.print("Enter Quantity: ");
        int qty = sc.nextInt();

        Product p = new Product(name, price);
```

```java
        Order o = new Order(p, qty);

        o.displayBill();

        sc.close();
    }
}
```

Output

```
Enter Product Name: Laptop
Enter Product Price: 50000
Enter Quantity: 2

Order Details
Product Name : Laptop
Price : 50000.0
Quantity : 2
Total Amount : Rs.100000.0

=== Code Execution Successful ===
```

9. Student Result Analyzer using Arrays of Objects

Program

```
import java.util.Scanner;

class Student {
    int roll;
    String name;
    int m1, m2, m3;

    Student(int roll, String name, int m1, int m2, int m3) {
        this.roll = roll;
        this.name = name;
        this.m1 = m1;
        this.m2 = m2;
        this.m3 = m3;
    }

    int total() {
        return m1 + m2 + m3;
    }

    double average() {
        return total() / 3.0;
    }

    String grade() {
        double avg = average();

        if (avg >= 90)
            return "A";
        else if (avg >= 75)
            return "B";
        else if (avg >= 50)
            return "C";
        else
            return "Fail";
    }

    void display() {
        System.out.println("\nRoll No : " + roll);
        System.out.println("Name : " + name);
        System.out.println("Total : " + total());
        System.out.println("Average : " + average());
        System.out.println("Grade : " + grade());
    }
}

public class Main {
    public static void main(String[] args) {
```

```java
        Scanner sc = new Scanner(System.in);
        Student[] s = new Student[3];

        for (int i = 0; i < 3; i++) {
            System.out.println("\nEnter Student " + (i + 1));

            System.out.print("Roll No: ");
            int roll = sc.nextInt();
            sc.nextLine();

            System.out.print("Name: ");
            String name = sc.nextLine();

            System.out.print("Marks in Subject 1: ");
            int m1 = sc.nextInt();

            System.out.print("Marks in Subject 2: ");
            int m2 = sc.nextInt();

            System.out.print("Marks in Subject 3: ");
            int m3 = sc.nextInt();

            s[i] = new Student(roll, name, m1, m2, m3);
        }

        int top = 0;

        System.out.println("\nStudent Results");

        for (int i = 0; i < 3; i++) {
            s[i].display();

            if (s[i].total() > s[top].total())
                top = i;
        }

        System.out.println("\nClass Topper");
        System.out.println("Name : " + s[top].name);
        System.out.println("Total Marks : " + s[top].total());

        sc.close();
    }
}
```

```
Enter Student 1
Roll No: 101
Name: aishu
Marks in Subject 1: 90
Marks in Subject 2: 95
Marks in Subject 3: 89

Enter Student 2
Roll No: 102
Name: abinaya
Marks in Subject 1: 80
Marks in Subject 2: 79
Marks in Subject 3: 81

Enter Student 3
Roll No: 103
Name: jasmine
Marks in Subject 1: 60
Marks in Subject 2: 56
Marks in Subject 3: 54
```

```
Student Results

Roll No : 101
Name : aishu
Total : 274
Average : 91.33333333333333
Grade : A

Roll No : 102
Name : abinaya
Total : 240
Average : 80.0
Grade : B

Roll No : 103
Name : jasmine
Total : 170
Average : 56.666666666666664
Grade : C

Class Topper
Name : aishu
Total Marks : 274
```

10. Smart Home Device Controller using Hierarchical Inheritance

Program

```
import java.util.Scanner;

class Device {
    String name;

    Device(String name) {
        this.name = name;
    }

    void turnOn() {
        System.out.println(name + " is ON");
    }

    void turnOff() {
        System.out.println(name + " is OFF");
    }

    void power() {
        System.out.println("Power Consumption : 0 W");
    }
}

class Light extends Device {
    Light() {
        super("Light");
    }

    void power() {
        System.out.println("Power Consumption : 20 W");
    }
}

class Fan extends Device {
    Fan() {
        super("Fan");
    }

    void power() {
        System.out.println("Power Consumption : 75 W");
    }
}

class AirConditioner extends Device {
    AirConditioner() {
        super("Air Conditioner");
    }
```

```java
    void power() {
        System.out.println("Power Consumption : 1500 W");
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Device[] d = new Device[3];

        d[0] = new Light();
        d[1] = new Fan();
        d[2] = new AirConditioner();

        System.out.println("Choose Device");
        System.out.println("1. Light");
        System.out.println("2. Fan");
        System.out.println("3. Air Conditioner");
        System.out.print("Enter Choice: ");

        int choice = sc.nextInt();

        if (choice >= 1 && choice <= 3) {
            d[choice - 1].turnOn();
            d[choice - 1].power();
            d[choice - 1].turnOff();
        } else {
            System.out.println("Invalid Choice");
        }

        sc.close();
    }
}
```

Output

```
Choose Device
1. Light
2. Fan
3. Air Conditioner
Enter Choice: 1
Light is ON
Power Consumption : 20 W
Light is OFF

=== Code Execution Successful ===
```